CY202 - Unit 5 Question Bank

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CY202: BIOLOGY AND ENVIRONMENTAL SCIENCE

Unit 5: Water Pollution Abatement and Solid Waste Management

QUESTION SHEET WITH MODEL ANSWERS

Section A - 2 Marks Questions | Section B - 7 Marks Questions

Compiled from the course syllabus, classroom notes and the teacher-provided question bank.

Syllabus coverage: water quality standards and parameters; sources of water pollution; water treatment - pre-, primary, secondary and advanced processes; COD and BOD of wastewater; solid waste - sources and integrated solid waste management; hazardous waste - sources and management; e-waste - source and management.

SECTION A - 2 MARKS QUESTIONS

Answer briefly, covering every topic of Unit 5.

A1. What are the three types of water quality parameters? Mention with examples.

Water quality is described by three classes of parameters. **(i) Physical** - temperature, turbidity, colour, solids (TDS/TSS) and electrical conductivity. **(ii) Chemical** - pH, dissolved oxygen (DO), biochemical oxygen demand (BOD), chemical oxygen demand (COD) and hardness. **(iii) Biological** - bacteria, algae, pathogens and faecal coliform.

A2. What do you mean by Total Dissolved Solids (TDS)? How can TDS be used to measure the ionic conductance of water?

TDS is the residue left when a filtered water sample is evaporated - that is, the solids that pass through the filter, while suspended solids are retained on it. Total solids, TS = TDS + TSS. TDS is related to electrical conductivity (EC) by the proportionality TDS (mg/L) = k_e x EC (µS/cm). Since an electric current is carried by ions in solution, conductivity increases as the ion concentration increases; therefore TDS is used to estimate the ionic strength (ionic conductance) of water, for example in groundwater recharge with treated wastewater. Water is classified by TDS as freshwater (<1500 mg/L), brackish (1500-5000 mg/L) and saline (>5000 mg/L).

A3. What is the saturation concentration of Dissolved Oxygen (DO) at 20°C and 0°C?

The saturation concentration of DO is about **9 mg/L at 20°C** and about **14.6 mg/L at 0°C**. DO is very sensitive to temperature - cold water holds more dissolved oxygen than warm water.

A4. Mention any two methods of determination of Dissolved Oxygen (DO) in water.

(i) The **Winkler titration (iodometric) method** - the traditional method; and (ii) the **electrometric method** (membrane electrode / DO meter). A third, quick and inexpensive option is the **colorimetric method**.

A5. The results from a BOD test, diluted by 100, are given in the table. Calculate BOD_5.

BOD_5 = p (DO_i - DO_f), where p = dilution factor (100) and DO_i, DO_f are the initial and final DO concentrations.

Time (days)	Diluted sample DO (mg/L)	BOD (calculated) (mg/L)
0	7.95	-
1	3.75	420
3	2.75	520
5	1.80	615

Therefore BOD_5 = 100 x (7.95 - 1.80) = **615 mg/L**.

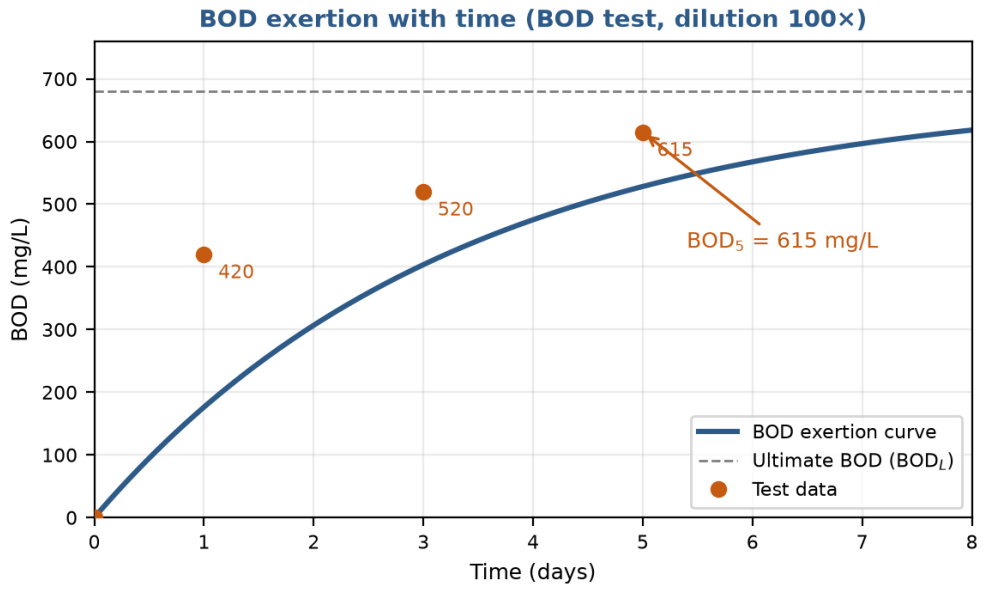


Fig. A5 - BOD exertion with time; BOD_5 is read at day 5.

A6. What is eutrophication?

Eutrophication is the process by which a lake changes from a clean, clear condition - with a relatively low concentration of dissolved nutrients and a balanced aquatic community - to a nutrient-rich, algae-filled state and thence to an oxygen-deficient, waste-filled condition. Excess nitrates and phosphates from sewage and fertiliser run-off promote algal blooms; when the algae die, microbial decomposition consumes dissolved oxygen, and anaerobic organisms then release gases such as methane and hydrogen sulphide. Eutrophication is a naturally occurring, slow and inevitable process, but when accelerated by human activity it is called cultural eutrophication.



Fig. A6 - Stages of eutrophication of a water body.

A7. What is activated sludge?

Activated sludge is the active, healthy suspension of biological solids - mostly bacteria - that forms in the aeration tank of the activated-sludge process when settled sewage is aerated in the presence of recirculated sludge. Under these oxygenated conditions the microorganisms thrive and metabolize the dissolved organic matter of the sewage, thereby lowering its BOD.

A8. Discuss in brief the working principle of a rotating biological contactor.

A rotating biological contactor (RBC) consists of a series of large plastic disks mounted on a horizontal shaft and partially submerged in primary effluent. As the shaft rotates, the disks are exposed alternately to air and to wastewater, so a layer (biofilm) of bacteria grows on the disks. The bacteria metabolize the organic matter in the wastewater, while the air exposure supplies the oxygen they need. It is a biological secondary-treatment method, less common than the trickling filter, activated sludge process or oxidation pond.

A9. Mention the goals of circular economy.

The goal of the circular economy (CE) is to optimize circle-system functions by reducing the resources exiting the system - that is, to minimise waste and to treat goods, materials and waste as secondary raw materials that can be recycled and reused. This sets it apart from the linear 'take-make-use-dispose' economy. Moving towards a greater circular economy helps meet the resource-efficiency goals of the strategy for smart, inclusive and sustainable growth; in wastewater treatment it promotes reuse of treated effluent for irrigation, industry, toilet flushing and groundwater replenishment.

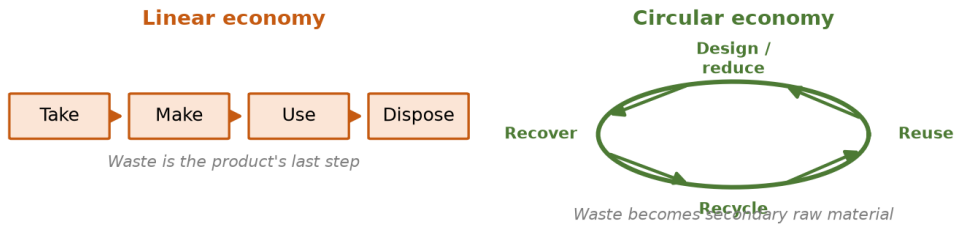


Fig. A9 - Linear economy versus circular economy.

A10. What are the principles of 3R?

The 3R principles are:

- **Reduce** - waste prevention (source reduction): cut unnecessary waste generation.
- **Reuse** - use products again, including reusable packaging, and refurbish goods to prolong life.
- **Recycle** - recover and reprocess materials into new products.

They form the core strategy of integrated solid waste management.

A11. Mention four benefits of e-waste recycling.

- **Conserves natural resources** - valuable metals (gold, silver, copper, palladium, aluminium) are recovered from old electronics, reducing the need to mine virgin material.
- **Protects the environment** - hazardous and toxic substances such as lead, mercury and cadmium are handled, processed and disposed of safely.
- **Creates jobs and economic value** - recycling supports professional recyclers and a secondary materials market.
- **Saves landfill space and reduces global warming** - less e-waste is dumped or incinerated, cutting greenhouse-gas (methane and CO_2) emissions and pollution of soil and water.

A12. Write any four effects of e-waste.

- **Air pollution** - burning e-waste releases hydrocarbons and toxic, carcinogenic substances into the atmosphere.
- **Soil contamination** - as e-waste breaks down it releases heavy metals such as lead, arsenic and cadmium, which enter plants and the human food supply.
- **Water pollution** - heavy metals (mercury, lithium, lead) from improper disposal leach from the soil into groundwater and surface water.
- **Human-health effects** - these toxins can cause birth defects, cancer and other health complications.

SECTION B - 7 MARKS QUESTIONS

Answer in detail with diagrams wherever applicable, covering every topic of Unit 5.

B1. What are the three phases of waste water treatment? Discuss, with diagrams, the working principle of the trickling filter.

Wastewater is treated in a sequence of phases, each removing a different class of impurity. Before the main phases, **preliminary treatment** (screening, grit removal, comminution) protects the plant's equipment and infrastructure.

- **Primary treatment** removes material that will either float or readily settle out by gravity - screening, comminution, grit removal and sedimentation in primary clarifiers (about two hours of detention time). The settled solids form raw or primary sludge; grease and floating matter are skimmed off mechanically.
- **Secondary treatment** removes the soluble organic matter that escapes primary treatment, using biological processes in which microbes consume the organic impurities as food and convert them to carbon dioxide, water and energy, thereby lowering the BOD. The three basic methods are the trickling filter, the activated sludge process and the oxidation pond; a fourth, less common method is the rotating biological contactor.
- **Tertiary (advanced) treatment** removes residual suspended solids, nutrients (nitrogen and phosphorus), toxic compounds and dissolved inorganics - for example effluent polishing by granular-media filters, nutrient removal, carbon adsorption and disinfection.

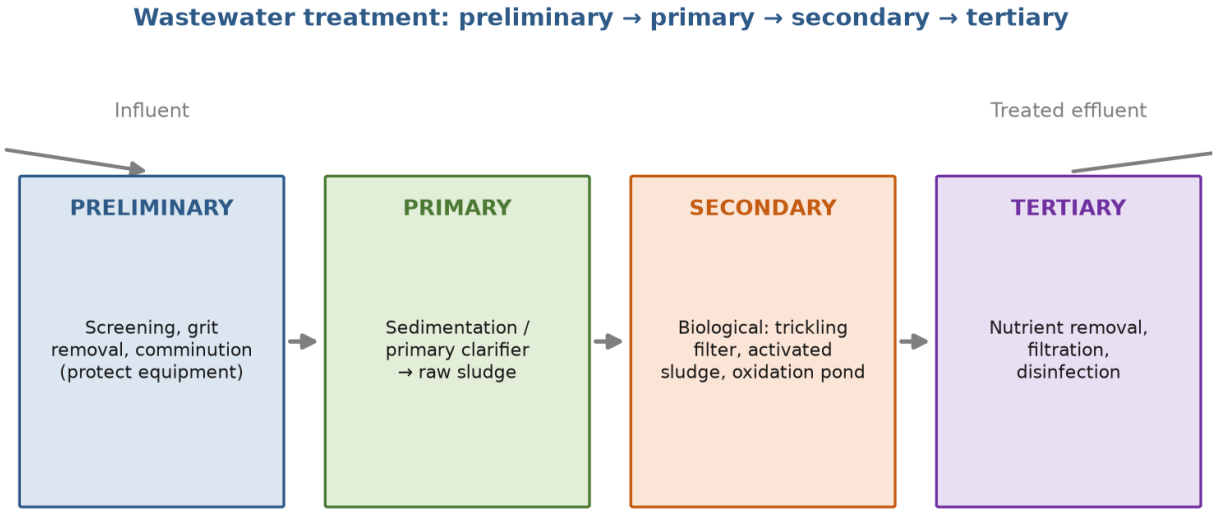


Fig. B1a - The phases of wastewater treatment.

Trickling filter - working principle. A trickling filter is simply a tank filled with a deep bed of stones. Settled sewage is sprayed continuously over the top of the stones and trickles to the bottom, where it is collected for further treatment. As the wastewater trickles down, bacteria gather and multiply on the stones, forming a biofilm. The steady flow of sewage over these growths allows the microbes to absorb the dissolved organics, thus lowering the BOD of the sewage. Air circulating upward through the spaces among the stones provides sufficient oxygen for the metabolic processes. Settling tanks called secondary clarifiers follow the filter and remove microbes washed off the stones; two or more filters may be connected in series, and sewage may be recirculated to increase treatment efficiency.

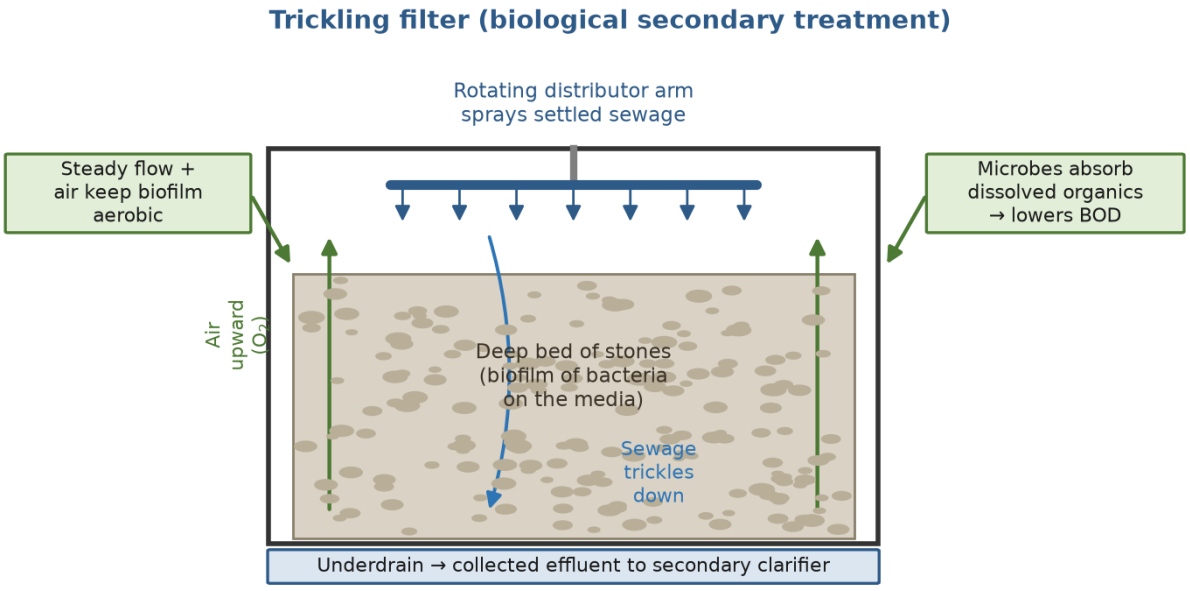


Fig. B1b - Cross-section of a trickling filter.

For comparison, in the **activated sludge process** settled sewage is mixed with recycled sludge in an aerated tank (about six hours' detention), where diffused air or mechanical mixers supply oxygen; the mixture then flows to a secondary clarifier, about 30 per cent of the settled sludge being returned to the aeration tank and the rest wasted.

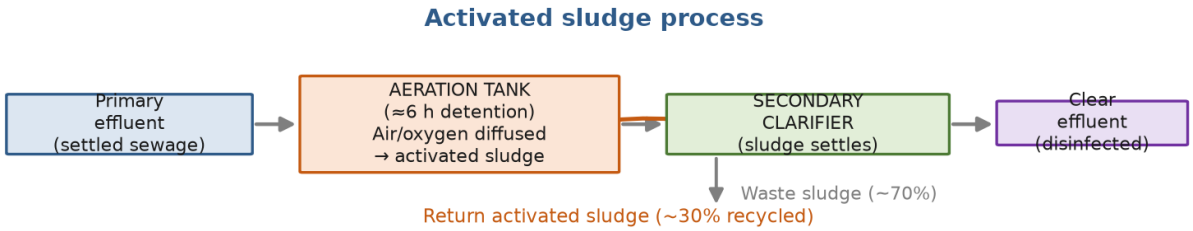


Fig. B1c - The activated sludge process (a second secondary-treatment method).

B2. What are biological oxidation ponds? Discuss their working principle with a proper diagram.

Oxidation ponds, also called lagoons or stabilization ponds, are large shallow ponds designed to treat wastewater through the interaction of sunlight, bacteria and algae. They are a biological secondary-treatment method.

Working principle. Algae grow using energy from the sun, carbon dioxide and the inorganic compounds released by bacteria in the water. During photosynthesis the algae release the oxygen needed by aerobic bacteria; the bacteria in turn oxidize the dissolved organic matter, releasing carbon dioxide and nutrients that the algae use. This symbiotic algae-bacteria cycle both stabilizes the organic waste and keeps the pond aerobic. Mechanical aerators are sometimes installed to supply additional oxygen, which reduces the required size of the pond. Sludge deposits in the pond must eventually be removed by dredging or scooping, and algae remaining in the effluent can be removed by filtration or by

a combination of chemical treatment and settling.

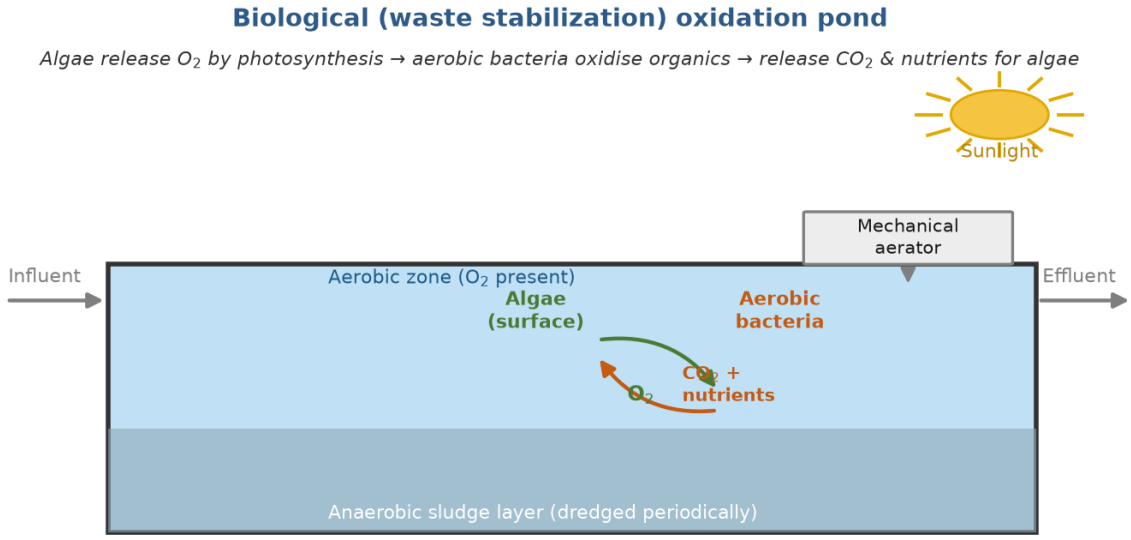


Fig. B2 - Cross-section of a biological oxidation (stabilization) pond.

B3. What is Integrated Solid Waste Management (ISWM)? Explain elaborately the four functional elements of ISWM.

Integrated solid waste management (ISWM) is the strategic approach to the sustainable management of solid wastes, covering all sources and all aspects - waste generation, segregation, transfer, sorting, treatment, recovery and disposal - in an integrated manner, with emphasis on maximising resource-use efficiency. It includes the collective management of all types of waste and implementation of the 3R (Reduce, Reuse, Recycle) policies. An efficient ISWM system evaluates local conditions and needs, then chooses, mixes and applies the most suitable activities. The four functional elements are:

1. **Source reduction (waste prevention)** - reduces unnecessary waste generation through products designed for recycling, durable and sustainable goods, reusable products and packaging, refurbishing goods to prolong product life, redesign to use less packaging, and reducing food spoilage. It lowers waste handling, transport and disposal costs and reduces methane generation.
2. **Recycling and composting** - recycling includes the accumulation, sorting and recovery of recyclables and their reprocessing into new products; composting converts organic waste into soil additives. Both create jobs, divert material from the waste stream and significantly reduce greenhouse-gas emissions.
3. **Waste transportation** - collection of waste from curbside and businesses, and from transfer stations where waste is concentrated and reloaded onto other vehicles for delivery to the landfill; it must be integrated systematically with the other activities.
4. **Waste disposal** - activities for waste that is not recycled, in particular sanitary landfilling and combustion. Landfills must be properly designed, well-constructed and systematically managed.

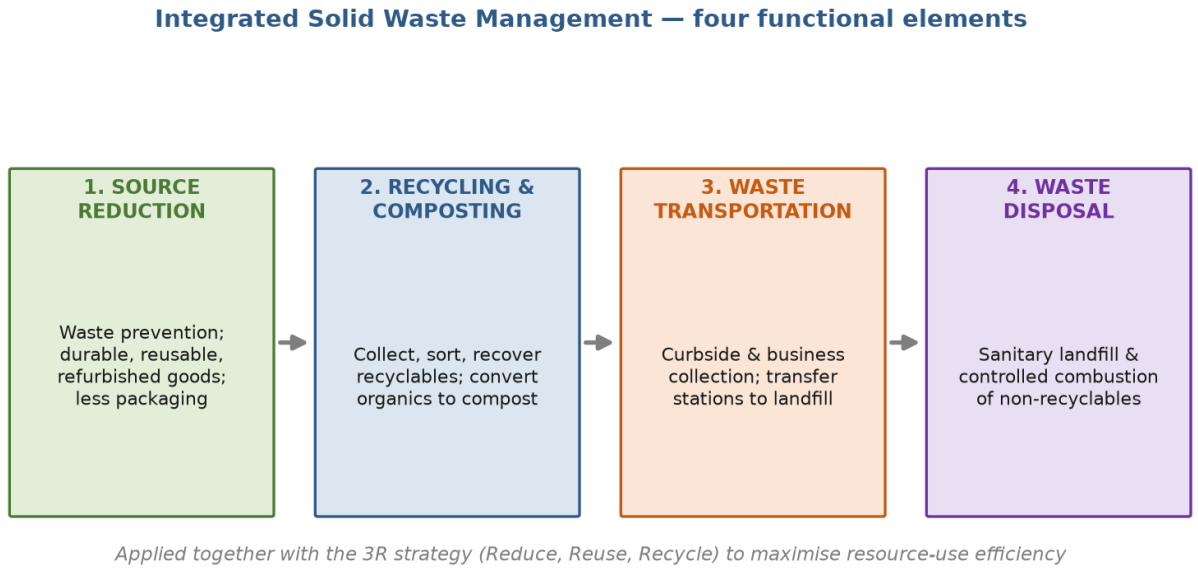


Fig. B3 - The four functional elements of integrated solid waste management.

Importance. With rapid population growth and economic development, waste generation keeps rising, putting pressure on society's ability to process and dispose of it. Inappropriately managed solid waste pollutes water, attracts rodents and insects, blocks drains (increasing floods) and raises greenhouse-gas emissions. A comprehensive ISWM system for efficient collection, transportation and systematic disposal - together with waste reduction and increased recycling - significantly reduces all these problems.

B4. Describe the major components of a hazardous waste management plan.

Hazardous waste is defined (USEPA) as any solid, liquid or gaseous substance which, by reason of any physical, chemical, reactive, toxic or infectious characteristics, causes danger to health or the environment when improperly treated, stored, transported or disposed of. Sources include industries, petroleum refineries, coke ovens, blast-furnace and scrubbing sludges, biological-waste laboratories, and thermal and nuclear power plants.

The main objective of a hazardous-waste management plan is to eliminate or reduce hazardous waste through process changes or resource recovery, and to account for the waste from its origin to ultimate disposal. Its major components are:

- **Inventory** - all industries producing more than 100 kg of hazardous waste per month must be registered, and an inventory of such generating agencies and the quantity produced per month prepared.
- **Storage** - generating plants must equip themselves with special on-site tanks, chemically resistant containers or basins for storage of waste.
- **Transport** - wastes must be transported by suitable tanker trucks, flat-bed trucks or rail cars to approved disposal sites.
- **Spillage** - a well-publicised emergency plan to deal with unexpected spillage or accidental release during transport, to prevent environmental damage and public-health hazard.
- **Disposal** - wastes transported to physical and chemical treatment plants for processing or concentration, or hauled directly to an approved hazardous-waste treatment facility for final disposal.
- **Waste exchange** - when on-site recycling or recovery is not feasible, transfer waste from large industries to smaller ones able to reuse low-purity oils, acids, alkalies or solvents, or recover valuable metals and materials.

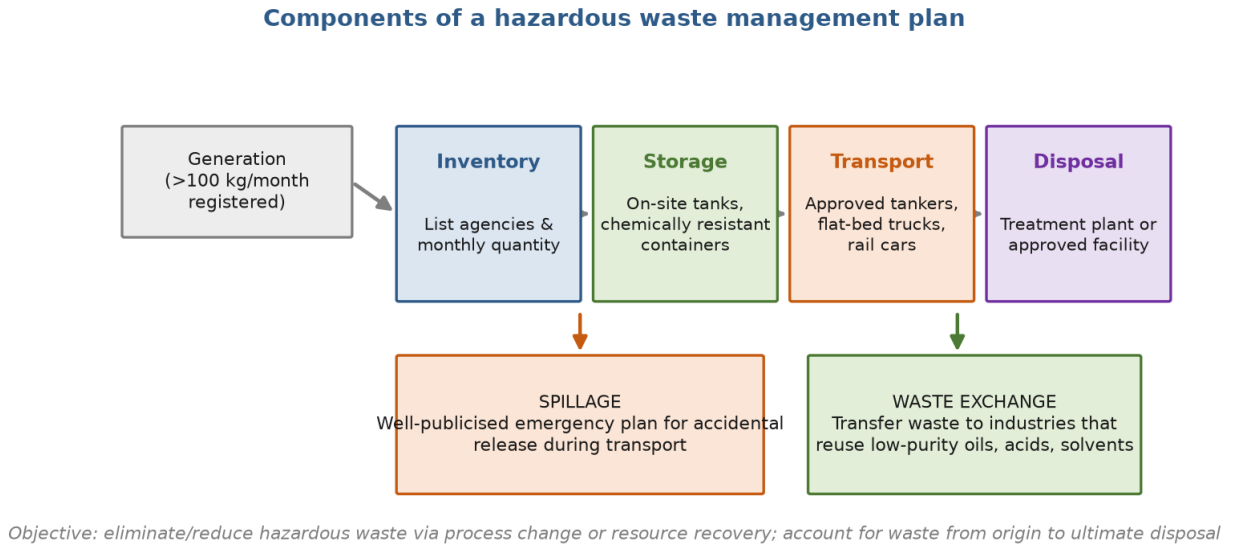


Fig. B4 - Major components of a hazardous waste management plan.

B5. What is e-waste? Cite their common sources and composition. Discuss the steps involved in e-waste recycling.

All items of electrical and electronic equipment and their parts that have been discarded by the user as waste, without the purpose of reuse or recycle, are called electronic waste (e-waste); stored used electronics are also considered e-waste. Improper processing of e-waste can lead to dangerous human-health effects and environmental pollution.

Sources. Large household appliances (refrigerators/freezers, washing machines, dishwashers, televisions); small household appliances (toasters, coffee makers, irons, hairdryers); IT and telecommunications equipment (personal computers, telephones, mobile phones, laptops, printers, scanners, photocopiers); lighting equipment (fluorescent lamps); electrical/electronic tools (handheld drills, saws, screwdrivers); toys and sports equipment; monitoring and control instruments; and automatic dispensers.

Composition. Valuable metals (gold, platinum, silver, palladium); useful metals (copper, aluminium, iron); hazardous substances (radioactive isotopes, mercury); toxic substances (dioxins); plastics (HIPS, ABS, PC, PPO); and glass (CRT glass of SiO_2, CaO, Na). A single mobile phone contains more than 40 elements.

Steps involved in e-waste recycling.

1. **Collecting and transporting** - recyclers place take-back booths or collection bins; when these fill, the e-waste is transported to recycling facilities.
2. **Shredding and sorting** - e-waste is broken into smaller pieces, then sorted and manually dismantled to retrieve different parts and separate core materials from components.
3. **Dust extraction** - the shredded particles are spread on a conveyor and broken down further; the dust is extracted and disposed of in an environmentally compliant manner.
4. **Magnetic separation** - a strong overhead magnet separates steel and iron from the other wastes.
5. **Water separation** - separates glass from plastic; lead-containing glass goes to smelters for batteries, X-ray tubes and new CRTs.
6. **Purification of the waste stream** - leftover metals are extracted from the plastics.
7. **Preparing recycled materials for sale** - the separated materials are prepared as raw materials for new electronics.



Fig. B5 - Steps in e-waste recycling.

B6. Discuss the benefits of e-waste recycling.

- **Conserves available natural resources** - recycling recovers valuable materials from old or unused electronic products, so manufacturers obtain raw materials from waste instead of mining copper, lead and other metals from nature.
- **Prioritises environmental protection** - hazardous and toxic substances such as lead, mercury and cadmium are handled, processed and managed safely, and dust from shredding is disposed of in an environmentally friendly manner, substantially reducing environmental hazards.
- **Creates jobs and economic value** - recycling creates new jobs for professional recyclers and a secondary market in recycled materials; in the US, recycling activities provided about 757,000 jobs, $6.7 billion in tax revenue and $36.6 billion in wages in a year (roughly 1.57 jobs, $76,000 in wages and $14,101 in tax revenue per 1,000 tons recycled).
- **Saves landfill space and reduces global warming** - recycling reduces the e-waste piling up at landfills and incinerators, which otherwise decompose to produce greenhouse gases (methane and CO_2) and pollute water and soil. Recycling a million laptops saves electric power equivalent to running 3,657 households for a year; a million cell phones yield about 75 lb of gold, 772 lb of silver, 35,274 lb of copper and 33 lb of palladium.

B7. What is Electromagnetic Radiation (EMR)? Mention its two types. Write the impacts of EMR exposure on human health. Explain in brief the mitigation techniques.

Electromagnetic radiation (EMR) is a fundamental form of energy that surrounds us everywhere - from the microwaves that heat our food to the radio waves that bring us music and news. With the rapid growth of mobile and wireless devices, telecom operators deploy many cellular towers that employ multiple antennas and transmit electromagnetic waves, so exposure to many EMF frequencies is increasing significantly. The EM field around an antenna is divided into regions - reactive near field, radiating near field (Fresnel) and radiating far field (Fraunhofer) - and RF-exposure assessment requires measuring the E-field, H-field and SAR (Specific Absorption Rate).

Two types of radiation:

- **Ionising radiation** - carries enough energy to break the chemical bonds of water and biological molecules.
- **Non-ionising radiation** - does not break these bonds, but is still harmful to human beings.

Impacts on human health. A high level of RF exposure heats body parts (tissues), because humans do not have strong heat-dissipating capability; parts such as the eyes and kneecaps are affected more. More severe effects can include behavioural abnormalities, cancer, physiological malfunctions (reproductive impairment, kidney failure), physical deformations and birth defects.

Mitigation techniques.

- **Reduce transmit power** - power density depends on the square of E/H, so lowering transmit power lowers the RF radiation level (though this reduces the coverage area).
- **Increase antenna height** - the RF field strength at the observer point falls as the distance increases.
- **Higher antenna gain with lower transmit power** can be used in close proximity to the antenna.
- **Basic health precautions** - minimise use of mobile and wireless devices, do not continuously wear wireless (Bluetooth/Zigbee) earphones, keep children's use low, and use devices in safer zones far from base stations or cellular towers.
- **Field management** - vertical positioning of current-carrying conductors on power-line structures produces lower magnetic fields; the electric-field component is easily shielded by most structures, though magnetic fields are difficult to contain. The ICNIRP has developed international guidelines and limits for EM radiation, and EM pollution can be tested with an ELF/EMF meter (a reading generally below 1.0 mG away from appliances is acceptable).

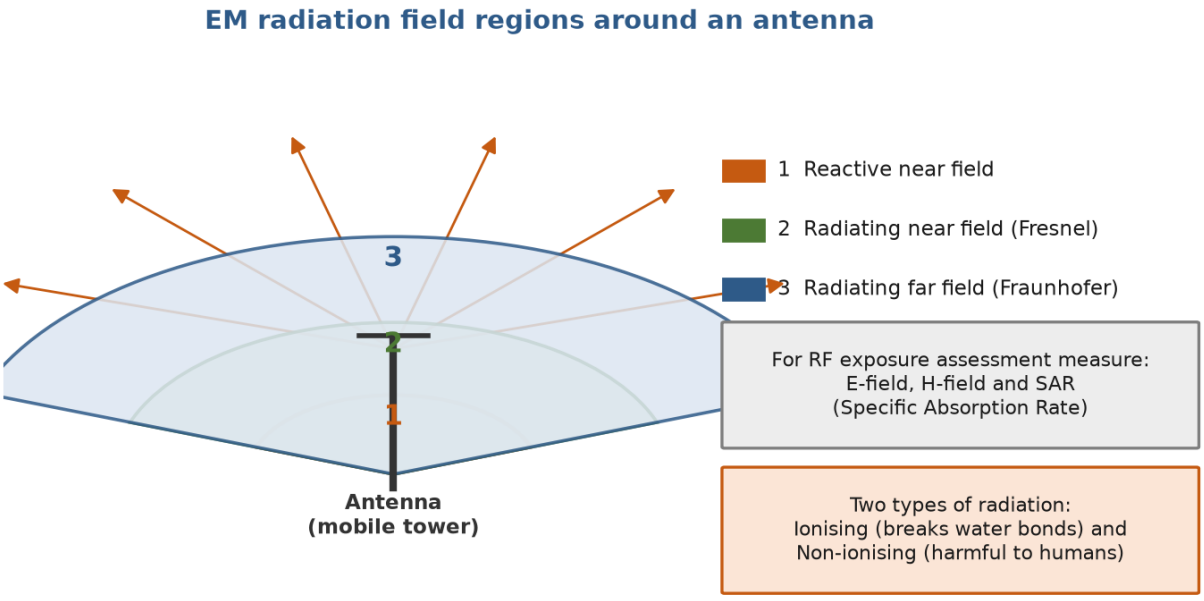


Fig. B7 - EM radiation field regions around an antenna.